Chapter: Matrices and Determinants

This document covers the comprehensive theoretical framework, operational properties, and analytical problem-solving techniques for Linear Algebra.

1. Foundations and Definitions

A matrix is a rectangular arrangement of elements in m rows and n columns. Matrices are fundamental to representing linear transformations.

Key Classifications:

- Square Matrix: Rows = Columns (n x n).
- Diagonal Matrix: All non-diagonal elements are 0.
- Identity Matrix (I): Diagonal elements are 1, others 0.

2. Determinants and Cofactor Expansion

The determinant is a scalar value calculated only for square matrices. It measures the "scaling factor" of a transformation.

det(A) = Σ a_ij * C_ij (Cofactor Expansion)

Properties: det(AB) = det(A)det(B), det(A^T) = det(A), and det(kA) = k^n det(A).

3. Matrix Inversion and Rank

Invertibility requires $\det(A) \neq 0$. The inverse is given by the adjoint method:

A^-1 = adj(A) / det(A)

Rank represents the number of linearly independent rows or columns.

4. Problem Bank & Solutions

Question 1: For matrix A = [[2, -1], [3, 1]], find A^-1.

Solution:
1. det(A) = (2*1) - (-1*3) = 2 + 3 = 5.
2. Adjoint of A: Swap diagonals (1, 2) and change sign of off-diagonals (1, -3).
3. adj(A) = [[1, 1], [-3, 2]].
4. A^-1 = 1/5 * [[1, 1], [-3, 2]] = [[0.2, 0.2], [-0.6, 0.4]].

Question 2: Evaluate the determinant of a 3x3 matrix where rows are linearly dependent.

Solution:
By property of determinants, if any row is a scalar multiple of another (linearly dependent), the determinant is **0**.

5. Advanced Theorems

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation: $\det(A - \lambda I) = 0$. This is highly effective for calculating high powers of matrices (e.g., A^5).